

Digital Logic Design

CSE - 260

Assignment 01

Summer 2025

Note that:

- 1. You will have to submit the hard copy, show all steps in all questions*
- 2. Make an cover page which contains you name, student ID and section*
- 3. You do not need to submit the ungraded portion.*

Deadline: 12 July,2025

Graded

1. Perform the following base conversions:
 - a. $(00110010101)_{1s} = (?)_9$
 - b. $(0011)_{BCD} = (?)_5$
 - c. $(101010101000011)_{1s} = (?)_{10}$
 - d. $(101010101000011)_{2s} = (?)_{10}$
 - e. $(101101100111.01011001)_{\text{Excess-4}} = (?)_{10}$
2. Suppose you have two 5-bit registers where your computer stores signed numbers. Now subtract 5 from 6 using your computer's registers, and figure out whether there is any overflow or not.
3. What is the smallest number you can store in a 8 bit register using a sign & magnitude system. Your final answer should be in HEX.

Ungraded

1. You are a computer engineer and you want to buy two 8 GB DDR4 RAMs. Each RAM costs $(1C2)_{16}$ dollars. You also want to buy a graphics card RTX4070Ti which costs $(10010110000)_2$ dollars. However, you don't have that much money with you and you are afraid to ask your parents about it. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you $(4064)_8$ dollars. How much will you have left after buying those components? (Show the answer in decimal)
2. Divide 43534212_5 by 124_5 . Find the quotient and remainder if any.
3. Multiply the following in the given base: $(34)_9$ with $(87)_9$. You must show the whole procedure.
4. Subtract -132 from -547 in 11 bits using 2's complement number system and justify whether there is an overflow or not.
5. Convert the following numbers to the equivalent given base:
 - a. $(79B.2FG)_{17} = (?)_{10}$
 - b. $(6045)_7 = (?)_{13}$
 - c. $(107.0425)_{10} = (?)_2$
6. Convert the following decimal numbers to 9 bits sign-and-magnitude, 1s complement, and 2s complement
 - a. $-(175)_{10}$
 - b. $+(123)_{10}$